$\tilde{\mathbf{E}}$ PHASOR TEMPLATES

Wavenumber per medium *nonmag.*

$$k_i = \frac{\omega\sqrt{\varepsilon_{r,i}}}{c}\ \text{(rad/m)} \quad \Rightarrow \quad k_2 = k_1\sqrt{\varepsilon_{r,2}/\varepsilon_{r,1}}$$

Direction table *$u \in \{x, y, z\}$ = axis perp. to boundary*

Inc dir	Inc	Refl	Trans
$+\hat{u}$	e^{-jk_1u}	e^{+jk_1u}	e^{-jk_2u}
$-\hat{u}$	e^{+jk_1u}	e^{-jk_1u}	e^{+jk_2u}

$+\hat{u}$ direction $\Leftrightarrow$ Med 2 at $u \geq 0$. Reflected sign flips, transmitted matches incident.

General template (incident has pol. $\hat{e}$, amplitude a_0):

$$\tilde{\mathbf{E}}^i = a_0\hat{e}\, e^{\mp jk_1u}$$

$$\tilde{\mathbf{E}}^r = \Gamma\, a_0\hat{e}\, e^{\pm jk_1u}$$

$$\tilde{\mathbf{E}}^t = \tau\, a_0\hat{e}\, e^{\mp jk_2u}$$

(lossy med 2: replace $e^{\mp jk_2u}$ with $e^{\mp\alpha_2u}e^{\mp j\beta_2u}$)

Total in medium 1: $\tilde{\mathbf{E}}_1 = \tilde{\mathbf{E}}^i + \tilde{\mathbf{E}}^r$.

$\hat{e}$ by polarization *plug into templates above*

Pol.	$\hat{e}$	Note
Lin. $\hat{x}$	$\hat{x}$	$E_y = 0$
Lin. $\hat{y}$	$\hat{y}$	$E_x = 0$
Lin. at 45°	$(\hat{x}+\hat{y})/\sqrt{2}$	in phase
RHC	$\hat{x} - j\hat{y}$	$\delta = -\pi/2$
LHC	$\hat{x} + j\hat{y}$	$\delta = +\pi/2$
General	$a_x\hat{x} + a_y e^{j\delta}\hat{y}$	elliptical

Γ, τ act on $\hat{e}$ unchanged. Only swap the polarization, not the formulas.

POWER REFLECTION & TRANSMISSION

% reflected *always*

$$\%P_r = 100\,|\Gamma|^2$$

% transmitted *η-ratio matters!*

$$\%P_t = 100\,|\tau|^2\,\frac{\eta_1}{\eta_2}$$

Check: $\%P_r + \%P_t = 100$.

If medium 2 is denser ($\eta_2 < \eta_1$), $|\tau| > 1$ but $\%P_t < 100\%$ because of the η_1/η_2 factor.

LOSSY MEDIA — 5 CASES

Loss tangent $\varepsilon' = \varepsilon_r\varepsilon_0$, $\varepsilon_0 \approx 1/(36\pi\times10^9)$

$$\frac{\varepsilon''}{\varepsilon'} = \frac{\sigma}{\omega\varepsilon_r\varepsilon_0}$$

Type	$\sigma/\omega\varepsilon$	$\eta_c,\ \alpha,\ \beta$
Lossless ($\sigma = 0$)	0	$\eta = \eta_0/\sqrt{\varepsilon_r}$ exact; $\alpha = 0$, $\beta = \omega\sqrt{\varepsilon_r}/c$
Low-loss	$< 10^{-2}$	$\eta_c \approx \frac{\eta_0}{\sqrt{\varepsilon_r}}\left(1+j\frac{\sigma}{2\omega\varepsilon'}\right)$; $\alpha \approx \frac{\sigma\eta_0}{2\sqrt{\varepsilon_r}}$, $\beta \approx \frac{\omega\sqrt{\varepsilon_r}}{c}$
Quasi-cond.	10^{-2}–10^2	exact: $(\eta/\sqrt{X})e^{j\theta_\eta}$, see below
Good cond.	$> 10^2$	$\eta_c \approx (1+j)\sqrt{\pi f\mu/\sigma}$; $\alpha \approx \beta \approx \sqrt{\pi f\mu\sigma}$
Magnetic	any	keep full $\sqrt{\mu_r/\varepsilon_r}\,\eta_0$

For Γ/τ: drop j correction; use $\eta_0/\sqrt{\varepsilon_r}$. Magnetic: keep μ_r.

Exact (quasi-cond) $X = \sqrt{1 + (\varepsilon''/\varepsilon')^2}$

$$\alpha = \omega\sqrt{\frac{\mu\varepsilon'}{2}}\,\sqrt{X-1}$$

$$\beta = \omega\sqrt{\frac{\mu\varepsilon'}{2}}\,\sqrt{X+1}$$

$$\theta_\eta = \tfrac{1}{2}\arctan\frac{\sigma}{\omega\varepsilon'}$$

SNELL'S LAW — OBLIQUE INCIDENCE

Single boundary *angles from normal*

$$\sin\theta_2 = \sin\theta_1\sqrt{\frac{\varepsilon_{r1}}{\varepsilon_{r2}}} = \sin\theta_1\,\frac{n_1}{n_2}$$

Multilayer chain *stack of slabs $1 \to 2 \to 3 \to 4$*

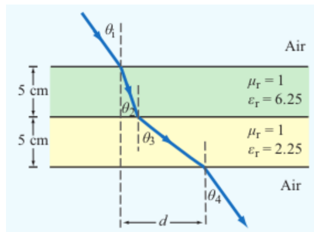
$$\sin\theta_{i+1} = \sin\theta_i\sqrt{\frac{\varepsilon_{r,i}}{\varepsilon_{r,i+1}}}$$

Air–slabs–air shortcut *outer media identical*

$$\theta_{\text{exit}} = \theta_{\text{incident}}$$

Snell only sets angles. Reflection/transmission magnitudes need Fresnel coefficients (parallel vs perpendicular).

LATERAL DISPLACEMENT



Beam offset through N slabs *thickness t_i, refracted angle inside slab i*

Slab-indexed *θ_i = angle inside slab i*

$$d = \sum_{i=1}^{N} t_i\,\tan\theta_i$$

Angle-indexed (HW3 P3 style) *θ_1 = incident, θ_{i+1} = inside slab i*

$$d = \sum_{i=1}^{N} t_i\,\tan\theta_{i+1}$$

Same physics, different numbering. In HW3 P3: slab 1 contributes $t_1\tan\theta_2$, slab 2 contributes $t_2\tan\theta_3$. The angle in the formula is always the one the ray makes inside *that slab.*

Air–slab–air: exit angle returns to θ_1 but is laterally shifted by d.

TABLE 8-2 — Γ, τ, R, T SUMMARY

***Med. 1** ($z < 0$) incident side; **Med. 2** ($z \geq 0$) transmitted. Shorthand: $c_i = \cos\theta_i$, $c_t = \cos\theta_t$.*

	Normal ($\theta_i = 0$)	$\perp$ pol	$\parallel$ pol
Γ	$\frac{\eta_2-\eta_1}{\eta_2+\eta_1}$	$\frac{\eta_2c_i-\eta_1c_t}{\eta_2c_i+\eta_1c_t}$	$\frac{\eta_2c_t-\eta_1c_i}{\eta_2c_t+\eta_1c_i}$
τ	$\frac{2\eta_2}{\eta_2+\eta_1}$	$\frac{2\eta_2c_i}{\eta_2c_i+\eta_1c_t}$	$\frac{2\eta_2c_i}{\eta_2c_t+\eta_1c_i}$
τ rel.	$\tau = 1+\Gamma$	$\tau_\perp = 1+\Gamma_\perp$	$\tau_\parallel = (1+\Gamma_\parallel)\frac{c_i}{c_t}$
R	$\lvert\Gamma\rvert^2$	$\lvert\Gamma_\perp\rvert^2$	$\lvert\Gamma_\parallel\rvert^2$
T	$\lvert\tau\rvert^2\frac{\eta_1}{\eta_2}$	$\lvert\tau_\perp\rvert^2\frac{\eta_1c_t}{\eta_2c_i}$	$\lvert\tau_\parallel\rvert^2\frac{\eta_1c_t}{\eta_2c_i}$
$R+T$	$T = 1-R$	$T_\perp = 1-R_\perp$	$T_\parallel = 1-R_\parallel$

Notes: $\sin\theta_t = \sqrt{\mu_1\varepsilon_1/\mu_2\varepsilon_2}\,\sin\theta_i$; *nonmag.* $\Rightarrow$ $\eta_2/\eta_1 = n_1/n_2$.

Intrinsic impedance η_i *plug into table above*

$$\eta_i = \sqrt{\frac{\mu_r}{\varepsilon_r}}\,\eta_0 \xrightarrow{\mu_r=1} \frac{\eta_0}{\sqrt{\varepsilon_{r,i}}}\ (\Omega) \qquad \eta_0 = 120\pi \approx 377\,\Omega$$

Default $\mu_r = 1$ (air, dielectric). Use $\mu_r \neq 1$ only if problem says "ferromagnetic"/iron/ferrite/etc.

Low-loss correction η_c $\sigma/(\omega\varepsilon) \ll 1$

$$\eta_c \approx \sqrt{\frac{\mu}{\varepsilon'}}\left(1 + j\frac{\sigma}{2\omega\varepsilon'}\right) \xrightarrow{\text{drop } j} \frac{\eta_0}{\sqrt{\varepsilon_r}}$$

For Γ/τ just use $\eta_0/\sqrt{\varepsilon_r}$. The j term is for $|\eta_c|, \theta_\eta$. See LOSSY MEDIA — 5 CASES for all regimes.

RECTANGULAR WAVEGUIDE — MODES

TE mode $E_z = 0,\ H_z \neq 0$

TM mode $H_z = 0,\ E_z \neq 0$

Dimensions $a \times b$ with $a > b$ along $\hat{x}, \hat{y}$; wave in $\hat{z}$.

E_x form (TE) *read m, n from arguments*

$$E_x \propto \cos\left(\frac{m\pi x}{a}\right)\sin\left(\frac{n\pi y}{b}\right)\sin(\omega t - \beta z)$$

coef. on $x = m\pi/a$; coef. on $y = n\pi/b$.

Identify TE$_{mn}$ vs TM$_{mn}$ *cos/sin pattern of E_x is identical for both!*

1. Read the problem statement (e.g. "a TE wave" $\to$ TE).
2. If $m = 0$ or $n = 0$ in the mode label $\Rightarrow$ must be TE (TM forbids 0).
3. Source field $H_z(x,y)$ given $\Rightarrow$ TE. Source field $E_z(x,y)$ given $\Rightarrow$ TM.

TE allows at least one of $m, n \geq 1$ (other can be 0). TM needs both $m, n \geq 1$. That's why TE$_{10}$ exists but TM$_{10}$ doesn't.

TABLE 8-3 — FULL FIELD COMPONENTS

Shorthand: $C_X = \cos\frac{m\pi x}{a}$, $S_X = \sin\frac{m\pi x}{a}$, $C_Y = \cos\frac{n\pi y}{b}$, $S_Y = \sin\frac{n\pi y}{b}$. *All fields carry an $e^{-j\beta z}$ factor (omitted below).* $k_c^2 = (m\pi/a)^2 + (n\pi/b)^2$.

TE mode *generator: $\tilde{H}_z$*

$$\tilde{H}_z = H_0\,C_X C_Y$$

$$\tilde{E}_x = \frac{j\omega\mu}{k_c^2}\left(\frac{n\pi}{b}\right)H_0\,C_X S_Y$$

$$\tilde{E}_y = -\frac{j\omega\mu}{k_c^2}\left(\frac{m\pi}{a}\right)H_0\,S_X C_Y$$

$$\tilde{E}_z = 0, \quad \tilde{H}_x = -\tilde{E}_y/Z_{\text{TE}}, \quad \tilde{H}_y = \tilde{E}_x/Z_{\text{TE}}$$

TM mode *generator: $\tilde{E}_z$*

$$\tilde{E}_z = E_0\,S_X S_Y$$

$$\tilde{E}_x = -\frac{j\beta}{k_c^2}\left(\frac{m\pi}{a}\right)E_0\,C_X S_Y$$

$$\tilde{E}_y = -\frac{j\beta}{k_c^2}\left(\frac{n\pi}{b}\right)E_0\,S_X C_Y$$

$$\tilde{H}_z = 0, \quad \tilde{H}_x = -\tilde{E}_y/Z_{\text{TM}}, \quad \tilde{H}_y = \tilde{E}_x/Z_{\text{TM}}$$

TEM (plane wave) for reference: $\tilde{E}_x = E_0e^{-j\beta z}$, $\tilde{H}_y = \tilde{E}_x/\eta$, f_c N/A, $k = \omega\sqrt{\mu\varepsilon}$.

CUTOFF FREQUENCY

f_{mn} *m, n integers ≥ 0 (not both zero)*

$$f_{mn} = \frac{u_{p0}}{2}\sqrt{\left(\frac{m}{a}\right)^2 + \left(\frac{n}{b}\right)^2}\ \text{(Hz)}$$

Common cutoffs ($a > b$)

Quantity	Formula
u_{p0}	$c/\sqrt{\varepsilon_r}$ (hollow: $u_{p0} = c$)
TE$_{10}$	$u_{p0}/(2a)$ (dominant)
TE$_{01}$	$u_{p0}/(2b)$
TE$_{20}$	$u_{p0}/a = 2f_{10}$
TE$_{11}$,TM$_{11}$	$\frac{u_{p0}}{2}\sqrt{1/a^2 + 1/b^2}$

Propagation requires $f > f_{mn}$.

ε_r FROM DISPERSION

Given ω, β, **mode** (m, n)

$$\varepsilon_r = \frac{c^2}{\omega^2}\left[\beta^2 + \left(\frac{m\pi}{a}\right)^2 + \left(\frac{n\pi}{b}\right)^2\right]$$

Sanity: β in rad/m, a, b in m, ω in rad/s.

WAVE IMPEDANCE & H_y

$$Z_{\text{TE}} = \frac{\eta}{\sqrt{1-(f_c/f)^2}}\ (\Omega) \qquad Z_{\text{TM}} = \eta\sqrt{1-(f_c/f)^2}$$

$$\eta = \frac{\eta_0}{\sqrt{\varepsilon_r}} \qquad H_y = \frac{E_x}{Z_{\text{TE}}}\ \text{(TE mode)}$$

$Z_{TE} > \eta$ always (denominator < 1); $Z_{TM} < \eta$.

GROUP VELOCITY & TRAVEL TIME

$$u_g = u_{p0}\sqrt{1-(f_{mn}/f)^2}\ \text{(m/s)}$$

$$u_p = \frac{u_{p0}}{\sqrt{1-(f_{mn}/f)^2}}\ \text{(m/s)} \quad \Rightarrow \quad u_pu_g = u_{p0}^2\ (\text{m}^2/\text{s}^2)$$

$$t = L/u_g\ \text{(s)} \quad \text{travel time through guide of length } L$$

Higher modes have lower u_g $\Rightarrow$ they arrive later. Group delay spread = sum over excited modes.

BOUNDARY

η_1, η_2 — imp. (Ω)
Γ — refl. coeff (unitless)
$\tau = 1 + \Gamma$ — trans. coeff
$k_i = \omega\sqrt{\varepsilon_{r,i}}/c$ (rad/m)
α_2, β_2 — if med 2 lossy
$z = 0$ — boundary plane
$\eta_0 = 120\pi \approx 377\,\Omega$

SNELL / OBLIQUE

θ_i, θ_t — angles from normal
$n_i = \sqrt{\varepsilon_{r,i}}$ — index of refr.
t_i — slab thickness (m)
d — lateral disp. (m)
plane of inc. = plane of $\hat{k}$+normal
$\perp$: $\mathbf{E} \perp$ this plane
$\parallel$: $\mathbf{E} \parallel$ this plane

WAVEGUIDE

a, b — wide, narrow dim. (m); $a > b$
m, n — mode indices (int.)
f_{mn} — cutoff (Hz)
β — prop. const (rad/m)
$Z_{\text{TE/TM}}$ — wave imp. (Ω)
E_x, H_y — field comps (V/m, A/m)
$\eta = \eta_0/\sqrt{\varepsilon_r}$

VELOCITIES

$u_{p0} = c/\sqrt{\varepsilon_r}$ — bulk phase vel.
$u_p = u_{p0}/\sqrt{1-(f_c/f)^2}$
$u_g = u_{p0}\sqrt{1-(f_c/f)^2}$
$u_pu_g = u_{p0}^2$
$t = L/u_g$ — travel time
$c = 3\times10^8$ m/s

POWER & UNITS

$\%P_r = 100|\Gamma|^2$
$\%P_t = 100|\tau|^2\eta_1/\eta_2$
$\%P_r + \%P_t = 100$
σ (S/m), ε (F/m), μ (H/m)
$\varepsilon_0 = 8.85\times10^{-12}$ F/m
$\mu_0 = 4\pi\times10^{-7}$ H/m
$1/(36\pi)\times10^{-9} \approx \varepsilon_0$